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SUPERSEDING
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MILITARY SPECIFICATION

WINTERIZATION: TRUCKS, COMMERCIAL

This specification is approved for use by US Army Tank-Automotive Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the winterization of commercial type trucks for military operations in an arctic environment. Winterization of arctic vehicles shall include necessary modifications, adjustments, and additions of componentary and materiels to the vehicle cab, engine, battery box, hood and accessories.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. Unless otherwise specified, the following specifications, standards, and handbooks of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation form a part of this specification to the extent specified herein.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: US Army Tank-Automotive Command, ATTN: AMSTA-GDS, Warren, MI 48397-5000, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document, or by letter.

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SPECIFICATIONS
FEDERAL

W-B-131	- Battery, Storage, Vehicular, Ignition, Lighting and Starting.
TT-C-490	- Cleaning Methods for Ferrous Surfaces and Pretreatments for Organic Coatings.
ZZ-H-428	- Hose, Non-Metallic and Hose; Preformed: (for the Coolant Systems of Automatic and other Liquid-Cooled Systems).
CCC-C-428	- Cloth, Duck, Cotton, Fire, Water, Weather and Mildew Resistant.

MILITARY

MIL-T-704	- Treatment and Painting of Materiel.
MIL-C-11755	- Antifreeze, Arctic-Type.
MIL-W-12332	- Welding, Resistance, Spot, Seam, and Projection, for Fabricating Assemblies of Low-Carbon Steel.
MIL-I-22023	- Insulation Felt, Thermal and Sound Absorbing.
MIL-C-46168	- Coating, Aliphatic Polyurethane, Chemical Agent Resistant.
MIL-P-52192	- Primer Coating, Epoxy.
MIL-P-53030	- Primer Coating, Epoxy, Water Reducible, Lead and Chromate Free.

STANDARDS
FEDERAL

FED-STD-595	- Colors.

MILITARY

MIL-STD-105	- Sampling Procedure Tables for Inspection by Attributes.
MIL-STD-130	- Identification Marking of U.S. Military Property.
MIL-STD-461	- Electromagnetic Emission and Susceptibility Requirements for the Control of Electromagnetic Interference.
MIL-STD-462	- Electromagnetic Interference Characteristics, Measurement of.
MIL-STD-1261	- Welding Procedures for Constructional Steels.
MIL-STD-1472	- Human Engineering Design Criteria for Military Systems, Equipment and Facilities.
MIL-STD-45662	- Calibration System Requirements.
MS33611	- Tube Bend Radii (ASG).
MS35000	- Battery, Storage, Lead-Acid, Waterproof.
MS35842	- Clamp, Hose, Low Pressure Type F.

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HANDBOOKS

MIL-HDBK-759	- Human Factors Engineering Design for Army Materiel.

2.1.2 Other Government documents, drawings, and publications. The following documents, drawings, and publications form a part of this specification to the extent specified herein.

DRAWINGS
ARMY

8383855	- Insulation, Thermal, Commercial.
10896643	- Thermal Barrier Material Insulation.
11668948	- Water Jacket, Complete.
11668950	- Swingfire Heater Assembly.

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Other publications. The following document(s) form a part of this specification to the extent specified herein. The issues of the documents which are indicated as DoD adopted shall be the issue listed in the current DoDISS and the supplement thereto, if applicable.

DEPARTMENT OF TRANSPORTATION
Federal Motor Vehicle Safety Standards.
Federal Motor Carrier Safety Regulations.

(Application for copies should be addressed to the Department of Transportation Federal Highway Administration, Washington, D.C. 20591.)

ENVIRONMENTAL PROTECTION AGENCY
Control of Air Pollution from New Motor Vehicles and New Motor Vehicle Engines.

(Application for copies should reference "Code of Federal Regulations to CFR and the Federal Register" and should be addressed to the Superintendent of Documents, US Government Printing Office, Washington, D.C. 20402.)

SOCIETY OF AUTOMOTIVE ENGINEERS, INC. (SAE)
SAE Standards and Recommended Practices

SAE J537	- Storage Batteries.
SAE J381	- Windshield Defrosting Systems Test Procedure - Trucks, Buses, and Multipurpose Vehicles.

(Application for copies should be addressed to the Society of Automotive Engineers, Incorporated, 400 Commonwealth Drive, Warrendale, Pennsylvania 15096.)

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(Technical Society and Technical Association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 First article. Unless otherwise specified (see 6.2), the contractor shall furnish one winterized vehicle which shall be subjected to first article inspection (see 4.4). First article inspection samples, properly marked with identifying information shall be representative of the vehicle to be furnished to the Government. All subsequent winterized vehicles delivered to the Government shall conform to these samples in all of their pertinent physical and performance attributes.

3.2 Materials. Materials not specified herein shall be of the quality regularly used by the manufacturer but shall be new and free from defects (see 4.8.1).

3.2.1 Dissimilar metals. When dissimilar metals are joined, at least one surface shall be pretreated with a material specified in MIL-T-704 which is appropriate for the end item application (see 4.8.1).

3.3 Design and construction. The winterized vehicles shall be capable of starting and operating in an ambient temperature of minus 65 degrees Fahrenheit (°F). Additions to winterized vehicles shall consist of the following (see 4.8.1):

a. A portable gasoline fired heater to preheat the engine coolant and the batteries; and to supply auxiliary heat to the cab.

b. Insulation for the vehicle cab, battery box and quilted covers for the hood.

3.3.1 Treatment and painting. Unless otherwise specified by the procuring activity for the appropriate service (see 6.2), the arctic vehicle accessories and components and any vehicle surfaces that are altered by the installation of the arctic components that make up the arctic vehicle shall be treated with chemical agent resistant coating (CARC) in accordance with MIL-C-46168; primed with MIL-P-52192 or MIL-P-53030 and painted in accordance with MIL-T-704 as specified in 3.3.1.1 (see 4.8.1).

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3.3.1.1 Exterior metal and plastic surfaces. All exterior metal and plastic surfaces shall be treated as specified in MIL-T-704, and finish painted with an aliphatic polyurethane chemical resistant paint having color, gloss and infrared reflectance properties of lusterless forest green (see 3.3.1) in accordance with MIL-C-46168. Metal surfaces shall be pretreated as specified in type I or type IV of TT-C-490. Surfaces shall be primed as specified in MIL-P-52192. When overcoated with the aliphatic polyurethane, there shall be sufficient intercoat adhesion to assure a good paint bond. The electrical accessories and components under the vehicle hood and in the vehicle cab may be manufacturer's standard color. All non-metallic surfaces and fabrics shall match lusterless green No. 34151 or FED-STD-595 (see 4.8.1).

3.3.1.1.1 Finishing of metals prior to paint treatment. All punched-out holes, drilled holes, tubing ends, bar stock, and sheet metal cut-off shall be deburred and edges blended to preclude sharp edges, torn fabrics and entrapments for foreign particles prior to treatment, painting, and marking (see 4.8.2).

3.3.2 Service and operating instructions. Service and operating instructions for the winterization equipment shall be furnished with each winterized vehicle (see 4.8.3).

3.3.3 Human engineering design criteria. The human engineering design criteria shall be in accordance with the applicable sections of MIL-STD-1472 and MIL-HDBK-759 (see 4.8.1).

3.3.4 Welding. Welding of ferrous metals, shall be in accordance with MIL-STD-1261 and MIL-W-12332 (see 4.8.1).

3.3.5 Bolts and nuts. Bolts and nuts shall be corrosion resistant. Threads on the bolts shall protrude not less than two full threads nor more than four full threads beyond the nut. Nuts shall be of the nylon-insert lock type (see 4.8.1).

3.3.6 Sheet metal screws. Sheet metal screws are not acceptable for attaching components or accessories. All fasteners shall be corrosion resistant (see 4.8.1).

3.3.7 Coolant hoses and hardware. The coolant system hoses, tubing, pipes and connectors required shall not impede the flow of preheated coolant. Hoses and tubing shall maintain the proper radius and shall be routed and clamped to preclude chafing and interface. Coolant hoses shall be in accordance with grade B, class 2 of ZZ-H-428, and shall be clamped in accordance with MS35842. All metal tubing used for the winterized vehicle shall be corrosion resistant (see 4.8.3).

3.3.8 Tires and tubes. Tube type tires shall be furnished and installed in accordance with the tire manufacturer's recommendations (see 4.8.4).

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3.3.9 Portable heater. The vehicle shall be equipped with the portable Eberspacken Swingfire heater (Espar Products Incorporated Drawing 11668950), which shall be used with specified adapters as both an engine preheater and as an auxiliary cab heater (see 4.8.3).

3.3.9.1 Engine preheater.

3.3.9.1.1 Heat exchanger. A heat exchanger shall be positioned to provide proper insertion of the portable heater to preheat the vehicle engine coolant and the battery(s). The water jacket shall be Eberspacker Water jacket of Espar Products Incorporated, Drawing 11668948. The outlet connection of the heat exchanger shall flow the heated coolant to the engine, while the inlet connection shall receive the coolant from the engine block to the heat exchanger. Connections to the inlet and outlet ports shall be determined by the convective thermal flow due to altitude and positioning of the heat exchanger (see 4.8.1).

3.3.9.1.1.1 Heat exchanger location. The heat exchanger shall be placed in a metal cabinet exterior to the vehicle. A heat exchanger drain shall be located so that when the drain plug is removed the engine coolant in the heat exchanger shall drain to the ground (see 4.8.1).

3.3.9.1.1.2 Engine preheater starter wiring system. A 12 volts (V) direct current (dc), starter wiring harness with a female receptacle on one end and the opposite end of the harness connected to the vehicle 12 V dc electrical system shall be furnished. The wiring harness shall extend from the vehicle 12 V dc electrical system to a receptacle (Espar Products, Incorporated Part Number 206-00085 or equal), and be compatible with the portable enigne heater starter connector (see 3.3.9.1.1.3). The receptacle shall be mounted on the exterior of the vehicle in an accessible and protective location, and furnished with a screwed on cover or a cap with safety chain, to preclude contamination. The wiring harness shall be secured to preclude excessive slack and chafing and shall be protected from heat, weather and damage (see 4.8.3).

3.3.9.1.1.3 Engine preheater starter cable. A starter cable, not less than 10 feet in length shall be provided, to plug into the receptacle specified in 3.3.9.1.1.2 and into the engine preheater plug connector (see 4.8.3).

3.3.9.1.1.4 Starting the engine preheater. The engine preheater shall be started by: plugging the starter cable into the plug connector (see 3.3.9.1.1.3), and inserting the starter cable into the receptacle (see 3.3.9.1.1.2). After starting the heater, disconnect the starter cable from the heater and from the receptacle, and insert the heater into the heat exchanger (see 3.3.9.1.1) to preheat the vehicle engine coolant and the battery(s). The portable heater shall be removed from the heat exchanger before starting the vehicle. A warning decal shall be placed on the vehicle instrument panel to at least depict the following: WARNING: "DO NOT OPERATE VEHICLE WHILE ENGINE PREHEATER IS IN OPERATION. DO NOT OCCUPY PERSONNEL AREA WHILE HEATER IS IN OPERATION." Contractor shall provide a plate depicting the procedure for starting and inserting the heater into the heat exchanger (see 4.8.3).

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3.3.9.2 Auxiliary cab heater.

3.3.9.2.1 Auxiliary heat adapter. An adapter shall be designed and located in a manner to provide the following functions (see 4.8.1).

a. To provide a mount for the auxiliary heater in the cab heating mode.
b. Provide a means to inject hot air into the cab.
c. Provide for combustion air and exhaust.
d. Minimize obstruction of vehicle usage.
e. Facilitate easy removal and installation of the auxiliary heater.

3.3.10 Vehicle components. Vehicle components shall be relocated and/or reworked as required to facilitate the installation of the heat exchanger and components. Headlights, parking lights and turn signal lights shall not be altered (see 4.8.1).

3.3.11 Coolant circulation system. The coolant circulation system shall be supported to accept the loads imposed on the coolant fittings, hoses and tubing. A bleed valve shall be provided to allow purging all air from the system to permit the flow of heated engine coolant (see 4.8.1 and 4.8.3).

3.3.12 Vehicle air cleaner intake. The engine air intake shall be baffled to preclude powdered snow from accumulating in the air intake (see 4.8.5).

3.3.13 Battery(s). Battery(s) shall either be 6TN in accordance with MS-35000, or maintenance free battery(s) in accordance with SAE J537 and W-B-131. Means shall be provided to maintain the battery(s) charge. Battery(s) installation shall conform to Federal Motor Carrier Safety Regulation 393.27 through 393.30 (see 4.8.1 and 4.8.3).

3.3.14 Storage, portable preheater. An enclosure shall be provided for storage of the portable preheater. The enclosure shall include a suitable rack, hasp and padlock. The padlock key shall be secured in the vehicle glove box or wired to the vehicle steering wheel. The enclosure shall be accessible to personnel, fire resistant and weatherproof. Stowage shall be in such a manner as to prevent fuel leakage (see 4.8.3).

3.3.15 Vehicle cab insulation. The vehicle cab interior ceiling, back panel, side panels and doors except around the handles, and the firewall shall be insulated. The floor shall be insulated between the sheet metal floor and the floormat. Insulation shall be in accordance with Drawing 10896643 and shall not creep or peel when subjected to minus 65°F to 125°F ambient temperatures (see 4.8.3).

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3.3.16 Heat exchanger/portable heater exhaust, hot air drench system. During the preheat operation, exhaust from the heat exchanger outlet port shall be routed to the battery box for hot air drenching the battery(s). The hot air within the battery box shall then be routed to atmosphere. Hot air flow shall not be impeded by excessive back pressure or leakage. The system shall be insulated and heat shielded, as required. The battery box interior shall be baffled to provide a uniform temperature of 150°F while the portable heater is operating in an ambient temperature of minus 65°F (see 4.8.1).

3.3.17 Battery box. The battery box shall provide a heat shield between the exterior wall and the battery(s) to prevent hot exhaust being circulated directly on the battery(s) case. The shield shall provide space at the bottom and top to allow the circulation of the hot exhaust gas under, over and around the battery(s). The box shall be provided with heat shields at the inlet port area to distribute the hot exhaust gas. The inlet port shall be near the bottom of the battery box. Battery(s) shall rest on risers. Risers shall be padded with acid resistant pads. Each battery shall be secured with a rectangular frame and secured individually to the battery box. The frame shall be coated with an acid-resistant coating to prevent battery damage. The battery box bottom shall be provided with four weep holes at each corner to drain condensation that may accumulate from the hot exhaust gases. The battery box shall be fitted with a cover that is crowned on the interior to preclude condensation dripping on the battery(s). The cover shall be secured and capable of removal with ease. A hasp for a 1 1/2-inch padlock shall be furnished. An exhaust pipe shall be furnished to exhaust the gas to the atmosphere. The exhaust pipe shall be fitted with a rain cap. Padlock keys shall be secured to the glove box or the steering wheel in an appropriate location (see 4.8.1).

3.3.18 Hot air system tubing and hardware. The hot air system shall incorporate flexible stainless steel tubing, type AISI-304. Adapters and clamps shall be provided to assure ease of system maintenance. The tubing shall be in accordance with the bend radii specified in MS33611. Tubing shall be insulated from the exhaust outlet of the heat exchanger to the battery compartment. The insulated exhaust tubing shall permits safe touching with bare hands. The exhaust tubing shall be secured to the exhaust of the heat exchanger and to the inlet of the battery box with suitable means to preclude loosening. Exhaust tubing shall be routed to preclude an adverse affect on vehicle components (see 4.8.1 and 4.8.3).

3.3.19 Quilted arctic covers. Hood and radiator quilted arctic covers shall be of cotton duck in accordance with type I, class 1 of CCC-C-428. The insulation shall be in accordance with type I, class 3 of MIL-I-22023 and drawing 8383855 of 8383851 - 8383858. The insulation shall be stitched between the inner and outer layers of the cotton duck, and shall be noncombustible. Covers shall be capable of accepting a 90 mile per hour (mph) wind load or a vehicle speed of 60 mph without evidence of deformation or failure of the fasteners or stitching. The cotton duck arctic quilted hood and radiator covers shall be as specified as 3.3.19.1. Radiator cover shall not obstruct the vehicle headlights, parking lights, or turn signals, and shall not obstruct the flow of air to the vehicle engine air cleaner. Flaps with fasteners shall be provided for any areas that may require service (see 4.8.1 and 4.8.3).

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3.3.19.1 Fasteners, hood and radiator cover. The fasteners for securing the quilted hood and radiator cover shall be of the thumb-button, one-quarter turn, tension spring hook type. Grommets, as specified, shall be stitched into the hem. The bottom of the radiator cover shall be secured with spring tensioners and hooks, and the top of the radiator cover shall be secured with straps and buckles to the hood cover. The sides of the radiator cover shall be secured with one-quarter turn fasteners as specified. Fasteners shall be corrosion resistant (see 4.8.1 and 4.8.3).

3.3.20 Servicing and adjusting. The contractor shall service and adjust the winterized components; brake system, filling and charging of the battery(s); inflation of all tires; complete lubrication with arctic lubricants when specified (see 6.2). The coolant system shall be serviced either with a solution of ethylene glycol and water in equal parts by volume, or arctic antifreeze MIL-C-11755 as specified (see 6.2); servicing of the windshield washer reservoir with water and additives; and visual inspection to verify that no necessary accessory or component will interfere with any other accessory or component during static and dynamic conditions, and that all fasteners are in place and torqued for their respective size (see 4.8.3).

3.3.21 Maintainability. Maintainability shall be provided for the following:

a. Dimensions of all access openings shall be suitable for a bare and arctic mitten hand.
b. Draining of coolant and fuel/lubricants shall be to the ground without draining on any part of the truck. When drains are located over other parts of the truck, integral tubes, pipes, troughs or other means shall be used to convey the coolant or fuel/lubricant from the drain to the ground. Drains shall be accessible without the removal of accessories or parts, other than opening covers (see 4.8.3).

3.3.22 Safety. All equipment and exposed portions of the equipment which are subject to extreme temperature; rotating or reciprocating parts which are by nature or location a hazard to personnel, shall be insulated, fully enclosed or properly guarded (see 4.8.3).

3.3.22.1 Federal Motor Vehicle Safety Standards. When approved vehicle components are required to be relocated or altered to accomodate the winterized components, the alterations shall comply with all applicable Federal Motor Vehicle Safety Standards in effect on the date of manufacture (see 4.8.1).

3.3.22.2 Flame resistance. Flame resistance shall be in accordance with Federal Motor Carrier Safety Regulation 393.84 and Federal Motor Vehicle Safety Standard 302 (see 4.8.1).

3.3.22.3 Battery box and cables. Battery box and cables shall be in accordance with the applicable portions of Federal Motor Carrier Safety Regulations 393.27, 393.28, 393.29, and 393.31 through 393.33 (see 4.8.1 and 4.8.3).

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3.4 Performance. The winterized vehicle shall be capable of starting, operating, and performing its mission in an ambient temperature as low as -65°F with relative humidity (rh) of 100 percent, at elevation up to 5 000 feet above sea level, and without benefit of solar radiation.

3.4.1 Vehicle cab heating. The vehicle cab interior heater shall maintain crew comfort at 40°F when the ambient outside temperature is minus 65°F and the vehicle is in operational condition (see 4.8.6).

3.4.2 Defrosting and de-icing. The vehicle defrosting system shall be capable of removing mist and frost from the vehicle windows, after the vehicle has been exposed to -65°F for 24 hours. Ice on the windshield shall be capable of being removed to provide a view from inside the vehicle to the outside in accordance with the illustrations depicted on SAE J381, within 60 minutes after the vehicle engine coolant system has been preheated, engine started and the operating temperature stabilized, and the vehicle engine operating at idle speed. The hot air blowers shall be of the two-speed type, individually controlled, and adjustable. The vehicle cab heater shall be operating and the controls set at maximum output (see 4.8.7).

3.4.3 Carbon monoxide and Nox concentration. The auxiliary heater exhaust system shall be installed in such a manner that will ensure that the concentration of NOx shall not exceed five parts per million (ppm), and the concentration of carbon monoxide shall not exceed 50 ppm (0.0050 percent) in the driver's compartment after eight hours of operation with all heaters operating the windows closed, and the vehicle engine at idle speed (see 4.8.8).

3.4.3.1 Air pollution control. The winterized vehicle components and accessories shall comply with the Environmental Protection Agency Regualtions governing control of air pollution from new motor vehicles and new motor vehicle engines in effect on the date of manufacture (see 4.8.1).

3.4.4 Radio interference suppression and electromagnetic interference (EMI). Winterization shall not degrade the radio interference and EMI suppression of the original vehicle.

3.4.4.1 Radio interference suppression. Unless otherwise specified (see 6.2), arctic components and accessories shall be suppressed to limit electromagnetic radiation in accordance with MIL-STD-461 (see 4.8.9.1).

3.4.4.2 Electromagnetic interference (EMI). When specified (see 6.2), electromagnetic interference (EMI) suppression shall be in accordance with MIL-STD-461 for class C1, group I vehicles and equipment. The contractor shall demonstrate that the operation of the installed arctic components and accessories meet the requirements specified herein and do not affect the overall vehicle electromagnetic interference characteristics (see 4.8.9.2).

3.5 Marking. Unless otherwise specified by the procuring activity for the appropriate service (see 6.2), marking shall be in accordance with MIL-STD-130 (see 4.8.2).

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s more than once in the same sample, or when the when the same defect occur

3.6 Workmanship. Workmanship of the winterization components shall ensure that all components used in the construction shall be free of sharp corners, cracks, burrs and defective parts which are harmful to handling and safety of personnel. All fabricated cover items shall be free of incomplete seams, torn fabrics, looseness of fasteners and fittings, distortion and foreign matter. The quality of the winterization components shall be in compliance with requirements with respect to material, measurements, protective treatment and identifications (see 4.8.2).

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order (see 6.2), the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform or witness any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Inspection equipment. Unless otherwise specified in the contract (see 6.2), the contractor is responsible for the provision and maintenance of all inspection equipment necessary to assure that supplies and services conform to contract requirements. Inspection equipment must be capable of repetitive measurements to an accuracy of 10 percent of the measurement tolerance. Calibration of inspection equipment shall be in accordance with MIL-STD-45662.

4.2 Classification of inspection:

a. First article inspection (see 4.4).
b. Quality conformance inspections (see 4.5).
   1. Examination (see 4.5.2).
   2. Acceptance tests (see 4.5.3).
c. Control tests (see 4.6).

4.3 Inspection conditions. Unless otherwise specified (see 6.2), all inspections shall be conducted under the following conditions:

a. Air temperature 23 $\pm$ 10°C
b. Barometric pressure 725 + 50mm Hg
   - 75
c. Relative humidity 50 $\pm$ 30 percent

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4.4 First article inspection. Unless otherwise specified (see 6.2), the Government shall select the first winterized vehicle produced under the production contract (see 6.2) for first article inspection. First article samples shall be inspected as specified in table I. Approval of the first article sample by the Government shall not relieve the contractor of his obligation to supply winterized vehicles that are fully representative of those inspected as a first article sample. Any changes or deviation of the production units from the first article sample shall be subject to the approval of the contracting officer.

4.4.1 First article inspection failure. Test item deficiencies during, or as a result of, the first article test, shall be cause for rejection of the items until evidence has been provided by the contractor that corrective action has been taken to eliminate the deficiency. Any deficiency found during, or as a result of the first article test, shall be evidence that all items already produced prior to completion of the first article test are similarly deficient unless contrary evidence satisfactory to the contracting officer is furnished by the contractor. Such deficiencies on all items shall be corrected by the contractor. The Government shall not accept products until first article testing is completed to the satisfaction of the Government.

TABLE I. Classification of inspections.

Title	Requirement	Inspection	First article	Quality conformance		
				Examination	Acceptance (100%)	Control
Material and construction	3.2 thru 3.2.1, 3.3, 3.3.1, 3.3.1.1, 3.3.3, thru 3.3.6, 3.3.9.1 thru 3.3.9.1.1.1, 3.3.9.2, 3.3.10, 3.3.11, 3.3.13, 3.3.16 thru 3.3.19.1, 3.3.22.1, 3.3.22.2, and 3.4.3.1	4.8.1	X			
Defects (see table II)	3.3.1.1.1, 3.5, and 3.6	4.8.2	X	X		
Features	3.3.2, 3.3.7, 3.3.9, 3.3.9.1.1.2, 3.3.9.1.1.3, 3.3.9.1.1.4, 3.3.11, 3.3.13, 3.3.14, 3.3.15, 3.3.18 thru 3.3.22, and 3.3.22.3	4.8.3	X		X	

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TABLE I. Classification of inspections - Continued.

Title	Requirement	Inspection	First article	Quality conformance		
				Examination	Acceptance (100%)	Control
Tubes and tires	3.3.8	4.8.4	X		X	
Air intake	3.3.12	4.8.5	X	X		
Vehicle cab heater	3.4.1	4.8.6	X		X	
Defrosting and de-icing	3.4.2	4.8.7	X		X	
Carbon monoxide and Nox concentration	3.4.3	4.8.8	X			X
Radio interference suppression	3.4.4.1	4.8.9	X			X
Electromagnetic interference	3.4.4.2	4.8.9.2	X			X

4.5 Quality conformance inspection.

4.5.1 Sampling.

4.5.1.1 Lot formation. An inspection lot shall consist of all the winterized vehicles of one type and part number, from an identifiable production period, from one manufacturer, submitted at one time for acceptance.

4.5.1.2 Sampling for examination. Samples for quality conformance examination shall be selected in accordance with general inspection level II of MIL-STD-105.

4.5.2 Quality conformance examinations.

4.5.2.1 Acceptable quality level. Each sample selected in accordance with 4.5.1.2 shall be examined to determine conformance to the following acceptable quality levels (AQL) on the basis of percent defective (see 6.4).

Classification	AQL
Major	1.0
Minor	2.5

4.5.2.2 Classification of defects. For examination purposes, defects shall be classified as listed in table II.

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specified. shall be verified to conform to the values

TABLE II. Classification of defects.

Category	Defect	Method of examination
Critical	None	
Major	AQL 1.0% Defective	
101	Assembly, incomplete (see 3.3).	Visual
102	Dimensions affecting interchangeability, out of tolerance (see 3.3.1 and 3.3.3).	SIE 1/
103	Identification marking, improper (see 3.5).	Visual
104	Faulty workmanship affecting performance (see 3.6).	Visual
Minor	AQL 2.5% Defective	
201	Dimensions not affecting interchangeability, out of tolerance (see 3.3.1 and 3.3.3).	SIE 1/
202	Faulty workmanship affecting appearance (see 3.6).	Visual

1/ SIE = Standard Inspection Equipment.

4.5.3 Acceptance tests (100 percent). Each winterized vehicle shall be subjected to the acceptance tests specified in table I.

4.6 Control tests. Control tests shall be conducted on one winterized vehicle from each lot of 100 units consecutively produced, except that not more than one test shall be performed in a six month period, nor less than one test, in a twelve month period. The winterized vehicle shall be subjected to the control tests specified in table I.

4.7 Failure. Failure of any winterized vehilce to pass any of the specified acceptance or control tests shall be cause for the Government to refuse acceptance of the production quantity represented, until action taken by the contractor to correct defects and prevent recurrence has been approved by the Government.

4.8 Methods of inspection.

4.8.1 Materials and construction. Conformance to 3.2 through 3.2.1, 3.3, 3.3.1, 3.3.1.1, 3.3.3 through 3.3.6, 3.3.9.1, 3.3.9.1.1.1, 3.3.9.2, 3.3.10, 3.3.11, 3.3.13, 3.3.16 through 3.3.19.1, 3.3.22.1, 3.3.22.2, and 3.4.3.1, shall be determined by inspection of contractor records providing proof or certification that design, construction, processing, and materials conform to requirements. Applicable records shall include drawings, specifications, design data, receiving inspection records, processing and quality control standards, vendor catalogs and certifications, industry standards, test reports, and rating data.

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4.8.2 Defects. Conformance to 3.3.1.1.1, 3.5 and 3.6, shall be determined by examination for the defects listed in table II. Examination shall be visual, tactile, or by measurement with standard inspection equipment.

4.8.3 Features. Conformance to 3.3.2, 3.3.7, 3.3.9, 3.3.9.1.1.2, 3.3.9.1.1.3, 3.3.9.1.1.4, 3.3.11, 3.3.13 through 3.3.15, 3.3.18 through 3.3.22, 3.3.22.3, shall be determined by exercising the winterized components, and by qualitative observation sufficient to demonstrate that specified characteristics and features are present and functional.

4.8.4 Tires and tubes. To determine conformance to 3.3.8, it shall be verified that the vehicle tires have tubes and that the manufacturer has provided written approval of the use of tubes for the tires and the use intended. The inflation shall be in accordance with the tire and rim specifications for the type, size, and ply rating of the tires.

4.6.5 Air intake. To determine conformance to 3.3.12, air cleaner baffling to prevent powdered snow from clogging the intake of the air horn or grill shall be verified.

4.8.6 Vehicle cab heavy-duty heater. To determine conformance to 3.4.1, the vehicle cab heavy-duty heater shall be demonstrated with instrumentation to show change in temperatures in the cab after the engine coolant has been stabilized and the cab heater has operated in increments of 10 minutes for one hour at an ambient temperature of -65°F.

4.8.7 Vehicle defrosting and de-icing. To determine conformance to 3.4.2, the ice on the exterior of the glass shall be melted electrically.

4.8.8 Carbon monoxide and NOx concentration test. To determine conformance to 3.4.3, carbon monoxide and NOx concentration shall be tested and certified when a fuel fired compartment heater is supplemented for a hot water coolant heater in the vehicle cab. The certification shall state the following:

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a. Humidity.
b. Ambient temperature.
c. Weather conditions.
d. Length vehicles operated:
    1. Hours.
    2. RPM.
e. All windows and doors closed.
f. Location of test equipment, make and model.

4.8.9 Radio interference suppression and electromagnetic interference.

4.8.9.1 Radio interference suppression. To determine conformance to 3.4.4.1, the contractor shall furnish with the vehicle descriptive data of all interference suppression devices used for the winterization of the vehicle.

4.8.9.2 Electromagnetic interference. To determine conformance to 3.3.4.2, the EMI shall be tested in accordance with the applicable provisions of MIL-STD-462 for the EMI devices used for the winterization of the vehicle.

4.8.10 Road test. Road test shall be performed with a simulated payload for 25 miles on unimproved roads at a maximum safe speed not to exceed 35 miles per hour (mph) and 75 miles on improved roads at speeds up to 55 mph. The total combined weight of the vehicle with payload and installed winterized components shall not exceed the gross vehicle weight rating of the vehicle.

4.8.11 Cold test. Unless otherwise specified (see 6.2), the winterized vehicle shall be tested at minus 65°F for 48 hours. The vehicle shall be tested to verify that the vehicle winterization components that were assembled to the vehicle will meet the requirements specified in this specification.

5. PACKAGING

5.1 Preservation, packaging, packing, and marking. Preservation, packaging, packing, and marking for the desired level shall be in accordance with the applicable packaging standard or packaging data sheet specified by the contracting authority (see 6.2).

6. NOTES

6.1 Intended use. The components and accessories covered by this specification are intended for use in winterizing all commercial type trucks for low temperature opreation of minus 65°F and in the arctic environment.

6.2 Ordering data. Acquisition documents should specify the following:

a. Title, number, and date of this specification.
b. Number and date of the applicable drawing.
c. Treatment and painting if other than specified (see 3.3.1).
d. Complete lubrication with arctic lubricants if required (see 3.3.20).

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e. Whether coolant shall be ethyl glycol water solution or arctic antifreeze (see 3.3.20).
f. Electromagnetic interference (EMI) if required (see 3.4.4.1).
g. Radio interference suppression if not required (see 3.4.4.2).
h. Cold room test, if not required (see 4.8.11).
i. Marking if other than specified (see 3.5).
j. If first article samples are not required (see 3.1).
k. If responsibility for inspection shall be other than as specified (see 4.1).
l. If responsibility for inspection equipment shall be other than as specified (see 4.1.1).
m. If inspection conditions shall be other than as specified (see 4.3).
n. If first article inspection is not required (see 4.4),
o. Selection of applicable level and packaging standard or packaging data sheet (see 5.1).

Custodian:
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Preparing activity:
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1. DOCUMENT NUMBER	2. DOCUMENT TITLE

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
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b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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